Week 5 Report: Feature Engineering, Augmentation, and Dimensionality Reduction

Overview

This week built on the Week 4 data: 1,608 training customers (234 responders) and 412 test customers. The work covered three steps:

- **Feature engineering:** five candidate features built from two or more fields, each tested before being kept.
- **Data augmentation:** SMOTE compared with two simpler ways of handling the small number of responders, plus a check on whether the dataset is too small.
- **Dimensionality reduction:** aggregation, a correlation filter, and PCA, compared on the same folds.

Everything learned from the data comes from the training set, synthetic rows are created only inside training folds, and the test set has not been used for any decision.

We kept two of five combined features, which raised cross-validated PR-AUC from 0.693 to 0.725 and improved it on all 15 folds. We tested synthetic data (SMOTE) but did not adopt it, and we reduced the feature set from 32 to 27 features with no loss in performance.

1. Feature engineering

For new features, we considered the following five, each built from two or more existing fields:

- **Household_Size:** 1 + Is_Partnered + Kidhome + Teenhome
- **Lives_Alone:** single (not partnered) with no children at home
- **Income_per_Person:** Income divided by household size
- **Catalog_Share:** catalog purchases as a share of total purchases
- **Deal_Share:** deal purchases as a share of total purchases

To make sure the household-based features made sense, we first checked that Kidhome and Teenhome were not counting the same child twice. Kidhome counts young children and Teenhome counts teenagers, so a child shouldn't appear in both, and the data supports this. The 302 training customers

with both young children and teens have a median age of 48, which falls between customers with only young children (36) and only teens (50). That fits families with one younger and one older child.

This check did turn up a different problem with Household_Size: it is an exact sum of Is_Partnered, Kidhome, and Teenhome, so it adds no new information to a linear model. Two Week 4 features have the same issue: Total_Purchases (the sum of web, catalog, and store purchases) and Total_Prev_Accepted (the sum of the five campaign flags).

We then wanted to see how much each candidate changed the PR-AUC score, to check whether it added a meaningful improvement. We added each feature on its own to the 30 Week 4 features and scored it on the same 15 cross-validation folds (see the table below).

Feature	Built from	Change in PR-AUC	Folds improved (of 15)	Decision
Income_per_Person	Income / (1 + Is_Partnered + Kidhome + Teenhome)	+0.033	15	Keep
Lives_Alone	Not partnered and no children at home (0/1)	+0.020	13	Keep
Deal_Share	NumDealsPurchases / Total_Purchases	+0.002	10	Drop: no real gain
Household_Size	1 + Is_Partnered + Kidhome + Teenhome	+0.000	9	Drop: exact sum of its parts
Catalog_Share	NumCatalogPurchases / Total_Purchases	-0.002	3	Drop: no gain

In the end, we kept two features: Income_per_Person and Lives_Alone. Together they raised PR-AUC from 0.693 to 0.725, an improvement on all 15 folds. We dropped Household_Size, Catalog_Share, and Deal_Share because they added little or nothing.

2. Data augmentation

Applying SMOTE within each training fold balanced the two classes. In the last fold, for example, it turned 188 real responders into 1,099 by adding 911 synthetic ones, growing the fold from 1,287 to 2,198 rows. Synthetic rows were never added to validation folds or the test set. Because SMOTE barely moved PR-AUC (0.725 with and without it) or ROC-AUC, we did not add any synthetic rows to the saved data.

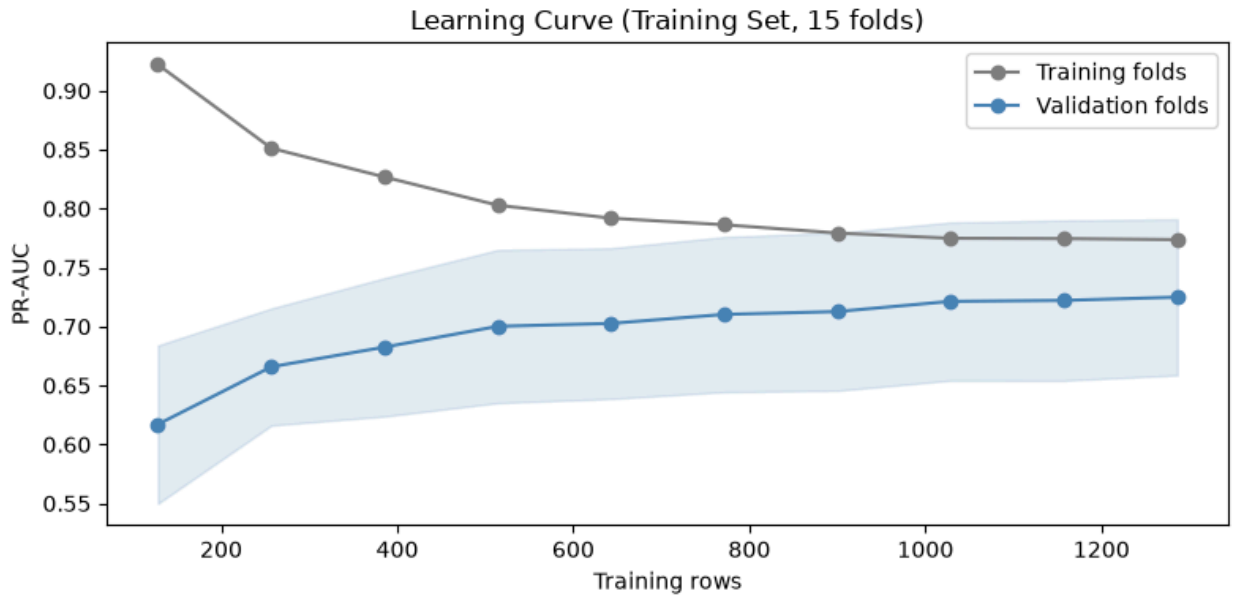
Approach	PR-AUC	ROC-AUC	Recall	Precision
No adjustment	0.725	0.929	0.543	0.747
Class weights	0.722	0.928	0.842	0.498
SMOTE synthetic data	0.725	0.927	0.839	0.520

Recall and precision use a 0.5 probability cutoff.

Is the dataset too small?

We also wanted to test whether our dataset is too small, given we only have 1,608 training customers and 234 responders. To do so, we built a learning curve: we trained the same logistic regression on growing shares of each training fold, from 10% to 100%, and scored each version on the same validation folds. If the scores were still climbing at full size, more data would help. After running it, the learning curve below shows validation PR-AUC rising quickly at first (0.617 with 128 rows, 0.700 with 514) and

then leveling off at 0.725, with the last 258 rows adding only 0.004. So we determined that sample size is not a major issue for this project, though it does limit how precisely we can measure results.



3. Dimensionality reduction

We also wanted to see whether we could reduce the number of features without losing predictive power. After feature engineering, the dataset had 32 features, and Week 3 had shown that several of them, such as income, spending, and purchase counts, are highly correlated. We tested three approaches, each compared with the full 32 features on the same 15 cross-validation folds.

We compared three approaches with the full 32 features on the same 15 folds:

1. **Aggregation:** drop AcceptedCmp1 to AcceptedCmp5 and keep their total, Total_Prev_Accepted, which has been in the data since Week 4.
2. **Correlation filter:** for each pair of features with a Spearman correlation of 0.85 or more, drop the one with the weaker link to Response. This removed Income, MntMeatProducts, MntWines, NumCatalogPurchases, and Total_Purchases.
3. **PCA:** replace all features with uncorrelated components, refit inside each fold. It takes 8 components to keep 80% of the variance, 13 for 90%, and 17 for 95%.

Feature set	Features	PR-AUC	Change vs. all	Folds matched or beat (of 15)
All features	32	0.725	-	-
1. Campaign flags replaced by total	27	0.729	+0.004	11
2. Correlation filter	27	0.711	-0.014	3
3. PCA, 90% of variance	13 components	0.702	-0.023	2
3. PCA, 95% of variance	17 components	0.702	-0.023	1

Aggregation was the only approach that reduced the feature count without hurting performance. It matched or beat the full set on 11 of 15 folds. This makes sense because the five flags carry little information that their total does not. Week 3 showed response rising steadily with the number of earlier campaigns accepted, from 8.1% with none to 79.2% with three or more, so the count is what matters.

The other two approaches lowered PR-AUC on most folds. The correlation filter removed Income and the top spending columns, which still carry some of their own signal even though they move together.

Based on these results, we chose the aggregated 27-feature set as our reduced set. With 32 features and 1,608 rows, dimensionality was not a major problem for this dataset. But the smaller set is simpler to explain and slightly less prone to overfitting with only 234 responders.

Model-ready data for Week 6

We saved two feature sets, both with real customers only and no missing values:

File pair	Features	Training rows	Test rows
train_model_full.csv / test_model_full.csv	32	1,608	412
train_model_reduced.csv / test_model_reduced.csv	27	1,608	412

The reduced set is the default. Keeping the full set lets Week 6 confirm the choice with stronger models than the yardstick used here.